

Nathan Chung

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science - GPA: 3.64/4.0

College Station, TX

Expected May 2028

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C/C++, C#

Frameworks/Libraries: FastAPI, Streamlit, SQLAlchemy, Pydantic, Plotly, React, Node.js, Next.js, TailwindCSS

Developer Tools: Git, Docker, GitHub Actions, pytest, ruff, MongoDB, REST APIs, WebSocket

AI/ML: pandas, scikit-learn, Claude API, OpenAI API, ChromaDB

EXPERIENCE

AMD (Advanced Micro Devices, Inc.)

Jan 2026 – Present

Software Engineer Intern

Austin, TX

- Built Python feature-validation scripts to configure and validate CPU power-management features for AMD's next-generation EPYC server platform (Venice), enabling repeatable regression testing across 10+ silicon features on remote dual-socket Linux systems.
- Identified and resolved critical bugs across multiple codebases and caught 8 incorrect register references via code review, migrated a register access library to fix dual-socket data misidentification, and debugged a silent write failure caused by an inverted control bit.
- Built dual-socket support for multi-IOD hardware configurations from scratch, expanding the team's validation coverage to span both single-IOD and multi-IOD system topologies.
- Built AI-powered Python tools to auto-generate test configurations from specifications and validate feature scripts for correctness, reducing manual setup across a distributed team spanning US, Malaysia, and India.

Arkane Digital

Jun 2025 – Aug 2025

Software Engineer Intern

Remote

- Developed and deployed 10+ AI-powered workflows on the Raikoo platform using Python and scikit-learn to automate portions of the front-end design-to-code pipeline.
- Built multi-component generation logic with text-preprocessing pipelines for the design-to-code system, supporting 20+ UI configurations across internal product teams.

PROJECTS

Python, FastAPI, Streamlit, SQLAlchemy, SQLite, pandas, scikit-learn, Docker

- Built a Python telemetry analysis tool (FastAPI + Streamlit + SQLAlchemy/SQLite) that ingests raw DAQ CSVs from hardware parameter sweeps and exposes 8 REST endpoints for sweep ingestion, comparison, time-series queries, and anomaly scoring
- Added ML-based run-quality scoring (scikit-learn Isolation Forest over 4 per-metric statistics) exposed via REST and surfaced as a sortable dashboard column that catches drift cases like normal average but high variance that static thresholds miss.
- Designed a normalized 3-table SQLite schema (sweeps, runs, metric_samples) with SQLAlchemy ORM to serve 4 distinct query patterns: ingestion, multi-run comparison, time-series extraction, and anomaly scoring without a heavyweight database dependency.

DocAgent | Python, FastAPI, async Postgres, pgvector, Docker, GitHub Actions

<https://huggingface.co/spaces/nathanwjchung/docagent>

- Built and shipped a full-stack RAG system on async FastAPI + Postgres /pgvector with Server-Sent Events streaming, multi-turn sessions, and Dockerized deploy to a public live demo on Hugging Face Spaces, 239 tests running against real Postgres in CI, sub-500ms first-token latency.
- Engineered an agent loop with tool-input validation against retrieved specs, so the model self-corrects on invalid calls instead of silently producing bad artifacts. Wired a 50 question evaluation suite as a GitHub Actions gate that fails PRs on quality regression, used to validate architecture swaps with measured deltas (cross-encoder reranking added +21 points to retrieval rank; 93% answer correctness).
- Architected around a pluggable domain-pack interface so the public portfolio published with zero domain-specific code in the 5,500-LOC generic core (enforced by a CI test scanning for forbidden tokens), same pattern lets a private internal domain ship without forking. .